

to estimate the summarized average treatment effect parameters discussed in Section 3.

Remark 9. *In applications with limited covariate overlap (i.e., with propensity scores sufficiently close to one), IPW and DR estimators may lead to imprecise (irregular) inference procedures, see, e.g., Khan and Tamer (2010). In such cases, provided that one is comfortable with (parametric) extrapolation and is sufficiently confident that the outcome regression working models are correctly specified, relying on the OR estimation approach may lead to more informative inferences. Alternatively, one may choose to trim extreme propensity score estimates though proceeding in this manner would change the target parameter; i.e., we would not be recovering the $ATT(g, t)$'s; see, e.g., Crump et al. (2009) and Yang and Ding (2018) for related discussion in other contexts. In the rest of the paper, we abstract from these points.*

4.1 Asymptotic Theory for Group-Time Average Treatment Effects

Next, we derive the asymptotic properties of our DR DiD estimators for the $ATT(g, t)$'s. To simplify exposition, we focus on the case with a never-treated comparison group as in (4.1); results that come from using the not-yet-treated group as the comparison group as in (4.2) follow from symmetric arguments and are therefore omitted. We also note that the theoretical results in this section are justified within the large n , fixed $\mathcal{T}$ paradigm.

Let $\|Z\| = \sqrt{\text{trace}(Z'Z)}$ denote the Euclidean norm of Z and set $W = (Y_1, \dots, Y_{\mathcal{T}}, X, D_1, \dots, D_{\mathcal{T}})$. For a generic $\kappa_{g,t}^{nev} = (\pi_g', \beta_{g,t,\delta}^{nev})'$, let

$$h_{g,t}^{dr,nev}(W; \kappa_{g,t}^{nev}, \delta) = (w_g^{treat}(W) - w_g^{comp,nev}(W; \pi_g)) (Y_t - Y_{g-\delta-1} - m_{g,t,\delta}^{nev}(X; \beta_{g,t,\delta}^{nev})),$$

where the normalized weights $w_g^{treat}(W)$ and $w_g^{comp,nev}(W; \pi_g)$ are given by

$$w_g^{treat}(W) = \frac{G_g}{\mathbb{E}[G_g]}, \quad w_g^{comp,nev}(W; \pi_g) = \frac{p_g(X; \pi_g) C}{1 - p_g(X; \pi_g)} \bigg/ \mathbb{E} \left[\frac{p_g(X; \pi_g) C}{1 - p_g(X; \pi_g)} \right]. \quad (4.3)$$

Let $g(\cdot)$ be a generic notation for $p_g(\cdot)$ and $m_{g,t,\delta}^{nev}(\cdot)$. With some abuse of notation, let $g(\cdot; \gamma)$ be a generic notation for $p_g(\cdot; \pi_g)$ and $m_{g,t,\delta}^{nev}(\cdot; \beta_{g,t,\delta}^{nev})$. The vector of pseudo-true parameters is given by $\kappa_{g,t}^{*,nev} = (\pi_g^{*t}, \beta_{g,t,\delta}^{*,nev})'$. Finally, let $\dot{h}_{g,t}^{dr,nev}(W; \kappa_{g,t}^{nev}) = \partial h_{g,t}^{dr,nev}(W; \kappa_{g,t}^{nev}) / \partial \kappa_{g,t}^{nev}$.

Assumption 7. (i) $g(x; \gamma)$ is a parametric model for $g(x)$, where $\gamma \in \Theta \subset \mathbb{R}^k$, Θ being compact; (ii) $g(X; \gamma)$ is a.s. continuous at each $\gamma \in \Theta$; (iii) there exists a unique pseudo-true parameter $\gamma^* \in \text{int}(\Theta)$; (iv) $g(X; \gamma)$ is a.s. twice continuously differentiable in a neighborhood of γ^* , $\Theta^* \subset \Theta$; (v) the estimator $\hat{\gamma}$ is strongly consistent for γ^* and satisfies the following linear expansion:

$$\sqrt{n}(\hat{\gamma} - \gamma^*) = \frac{1}{\sqrt{n}} \sum_{i=1}^n l_{g,t}(W_i; \gamma^*) + o_p(1),$$

where $l_{g,t}(\cdot; \gamma)$ is a $k \times 1$ vector such that $\mathbb{E}[l_{g,t}(W; \gamma^*)] = 0$, $\mathbb{E}[l_{g,t}(W; \gamma^*) l_{g,t}(W; \gamma^*)']$ exists and is positive definite and $\lim_{s \rightarrow 0} \mathbb{E} \left[\sup_{\gamma \in \Theta^*: \|\gamma - \gamma^*\| \leq s} \|l_g(W; \gamma) - l_g(W; \gamma^*)\|^2 \right] = 0$. In addition, (vi) for some $\varepsilon > 0$ and all $g \in \mathcal{G}$, $0 \leq p_g(X; \pi_g) \leq 1 - \varepsilon$ a.s., for all $\pi \in \text{int}(\Theta^{ps})$, where Θ^{ps} denotes the parameter space of π_g .

Assumption 8. For each $g \in \mathcal{G}$ and $t = \{2, \dots, \mathcal{T} - \delta\}$, assume that $\mathbb{E} \left[\|h_{g,t}^{nev}(W; \kappa_{g,t}^{*,nev}, \delta)\|^2 \right] < \infty$ and $\mathbb{E} \left[\sup_{\kappa \in \Gamma^*} |h_{g,t}^{nev}(W; \kappa)| \right] < \infty$, where Γ^* is a small neighborhood of $\kappa_{g,t}^{*,nev}$.

Assumptions 7-8 are standard in the literature, see e.g. Abadie (2005), Wooldridge (2007), Bonhomme and Sauder (2011), Graham et al. (2012), and Sant'Anna and Zhao (2020). Assumption 7 requires that the first-step estimators are based on smooth parametric models and that the estimated parameters admit $\sqrt{n}$ -asymptotically linear representations, whereas Assumption 8 imposes some weak integrability conditions. Under mild moment conditions, these requirements are fulfilled when one adopts linear/nonlinear outcome regressions or logit/probit models, for example, and estimates the unknown parameters by (nonlinear) least squares, quasi-maximum likelihood, or other alternative estimation methods, see e.g. Chapter 5 in van der Vaart (1998), Wooldridge (2007), Graham et al. (2012) and Sant'Anna and Zhao (2020). In other words, Assumptions 7-8 allow for flexible parametric specifications of the nuisance functions and accommodate different estimation methods.

In what follows, we write $w_g^{treat} = w_g^{treat}(W)$, $w_g^{comp}(\pi_g) = w_g^{comp,nev}(W; \pi_g)$, and $m_{g,t,\delta}^{nev}(\beta_{g,t,\delta}^{nev}) = m_{g,t,\delta}^{nev}(X; \beta_{g,t,\delta}^{nev})$ to minimize notation. For a generic $\kappa_{g,t}^{nev} = (\pi_g', \beta_{g,t,\delta}^{nev'})'$, define

$$\psi_{g,t,\delta}^{dr,nev}(W_i; \kappa_{g,t}^{nev}) = \psi_{g,t,\delta}^{treat,nev}(W_i; \beta_{g,t,\delta}^{nev}) - \psi_{g,t,\delta}^{comp,nev}(W_i; \pi_g, \beta_{g,t,\delta}^{nev}) - \psi_{g,t,\delta}^{est,nev}(W_i; \pi_g, \beta_{g,t,\delta}^{nev}), \quad (4.4)$$

with

$$\begin{aligned} \psi_{g,t,\delta}^{treat,nev}(W; \beta_{g,t,\delta}^{nev}) &= w_g^{treat} \cdot (Y_t - Y_{g-\delta-1} - m_{g,t,\delta}^{nev}(\beta_{g,t,\delta}^{nev})) \\ &\quad - w_g^{treat} \cdot \mathbb{E} [w_g^{treat} \cdot (Y_t - Y_{g-\delta-1} - m_{g,t,\delta}^{nev}(\beta_{g,t,\delta}^{nev}))], \\ \psi_{g,t,\delta}^{comp,nev}(W; \pi_g, \beta_{g,t,\delta}^{nev}) &= w_g^{comp}(\pi_g) \cdot (Y_t - Y_{g-\delta-1} - m_{g,t,\delta}^{nev}(\beta_{g,t,\delta}^{nev})) \\ &\quad - w_g^{comp}(\pi_g) \cdot \mathbb{E} [w_g^{comp}(\pi_g) \cdot (Y_t - Y_{g-\delta-1} - m_{g,t,\delta}^{nev}(\beta_{g,t,\delta}^{nev}))], \end{aligned}$$

and

$$\psi_{g,t}^{est,nev}(W; \pi_g, \beta_{g,t,\delta}^{nev}) = l_{g,t}^{or,nev}(\beta_{g,t,\delta}^{nev})' \cdot M_{g,t,\delta}^{dr,nev,1} + l_g^{ps,nev}(\pi_g)' \cdot M_{g,t,\delta}^{dr,nev,2},$$

where $l_{g,t}^{or,nev}(\cdot)$ is the asymptotic linear representation of the estimator for the outcome evolution of the comparison groups as described in Assumption 7(iv), $l_g^{ps,nev}(\cdot)$ is defined analogously for the generalized propensity score, and

$$\begin{aligned} M_{g,t,\delta}^{dr,nev,1} &= \mathbb{E} [(w_g^{treat} - w_g^{comp}(\pi_g)) \cdot \dot{m}_{g,t,\delta}^{nev}(\beta_{g,t,\delta}^{nev})], \\ M_{g,t,\delta}^{dr,nev,2} &= \mathbb{E} [\alpha_g^{ps,nev}(\pi_g) \cdot (Y_t - Y_{g-\delta-1} - m_{g,t,\delta}^{nev}(\beta_{g,t,\delta}^{nev})) \cdot \dot{p}_g(\pi_g)] \\ &\quad - \mathbb{E} [\alpha_g^{ps,nev}(\pi_g) \cdot w_g^{comp}(\pi_g) \cdot (Y_t - Y_{g-\delta-1} - m_{g,t,\delta}^{nev}(\beta_{g,t,\delta}^{nev})) \cdot \dot{p}_g(\pi_g)], \end{aligned}$$

with $\dot{m}_{g,t,\delta}^{nev}(\beta_{g,t,\delta}^{nev}) = \partial m_{g,t,\delta}^{nev}(X; \beta_{g,t,\delta}^{nev}) / \partial \beta_{g,t,\delta}^{nev}$, $\dot{p}_g(\pi_g) = \partial p_g(X; \pi_g) / \partial \pi_g$, and

$$\alpha_g^{ps,nev}(\pi_g) = \frac{C}{(1 - p_g(X; \pi_g))^2} \bigg/ \mathbb{E} \left[\frac{p_g(X; \pi_g) C}{1 - p_g(X; \pi_g)} \right].$$

Finally, let $ATT_{t \geq (g-\delta)}$ and $\widehat{ATT}_{t \geq (g-\delta)}^{dr,nev}$ denote the vector of $ATT(g, t)$ and $\widehat{ATT}_{dr}^{nev}(g, t; \delta)$, respec-

tively, for all $g \in \mathcal{G}_\delta$, $t \in \{2, \dots, \mathcal{T} - \delta\}$ such that $t \geq g - \delta$. Analogously, let $\Psi_{t \geq (g-\delta)}^{dr,nev}$ denote the collection of $\psi_{g,t,\delta}^{dr,nev}$ across all $g \in \mathcal{G}_\delta$, $t \in \{2, \dots, \mathcal{T} - \delta\}$ such that $t \geq g - \delta$. Consider the following claim:

$$\begin{aligned} & \text{For each } g \in \mathcal{G}_\delta, t \in \{2, \dots, \mathcal{T} - \delta\} \text{ such that } t \geq g - \delta, \\ & \exists \pi_g^* \in \Theta^{ps} : P(p_g(X; \pi_g^*) = p_g(X)) = 1 \quad \text{or} \\ & \exists \beta_{g,t,\delta}^{*,nev} \in \Theta^{reg} : P\left(m_{g,t,\delta}^{nev}(X; \beta_{g,t,\delta}^{*,nev}) = m_{g,t,\delta}^{nev}(X)\right) = 1. \end{aligned} \tag{4.5}$$

Claim (4.5) says that either the working parametric model for the generalized propensity score is correctly specified, or the working outcome regression model for the comparison group is correctly specified.

The next theorem establishes the joint limiting distribution of $\widehat{ATT}_{t \geq (g-\delta)}^{dr,nev}$.

Theorem 2. *Under Assumptions 1-4, 6-8, for each g and t such that $g \in \mathcal{G}_\delta$, $t \in \{2, \dots, \mathcal{T} - \delta\}$ and $t \geq g - \delta$, provided that (4.5) is true,*

$$\sqrt{n}(\widehat{ATT}_{dr}^{nev}(g, t; \delta) - ATT(g, t)) = \frac{1}{\sqrt{n}} \sum_{i=1}^n \psi_{g,t,\delta}^{dr,nev}(W_i; \kappa_{g,t}^{*,nev}) + o_p(1).$$

Furthermore, as $n \rightarrow \infty$,

$$\sqrt{n}(\widehat{ATT}_{t \geq (g-\delta)}^{dr,nev} - ATT_{t \geq (g-\delta)}) \xrightarrow{d} N(0, \Sigma)$$

where $\Sigma = \mathbb{E}[\Psi_{t \geq (g-\delta)}^{dr,nev}(W) \Psi_{t \geq (g-\delta)}^{dr,nev}(W)']$.

Theorem 2 provides the influence function for estimating the vector of group-time average treatment effects, $ATT_{t \geq (g-\delta)}$, as well as its limiting distribution. Importantly, Theorem 2 emphasizes the DR property of $\widehat{ATT}_{dr}^{nev}(g, t; \delta)$: it recovers the $ATT(g, t)$ provided that either the propensity score working model or outcome regression working model for the “never treated” is correctly specified.

In order to conduct inference, one can show that the sample analogue of Σ is a consistent estimator for Σ , which leads directly to standard errors and pointwise confidence intervals. Instead of following this route, we propose to use a simple multiplier bootstrap procedure to conduct asymptotically valid inference. Our proposed bootstrap leverages the asymptotic linear representations derived in Theorem 2 and inherits important advantages. First, it is easy to implement and very fast to compute. Each bootstrap iteration simply amounts to “perturbing” the influence function by a random weight V , and it does not require re-estimating the propensity score in each bootstrap draw. Second, in each bootstrap iteration, there are always observations from each group. This can be a real problem with the traditional empirical bootstrap where there may be no observations from a particular group in some particular bootstrap iteration. Third, computation of simultaneously (in g and t) valid confidence bands is relatively straightforward. This is particularly important since researchers are likely to use confidence bands to visualize estimation uncertainty about $ATT(g, t)$. Unlike pointwise confidence bands, simultaneous confidence bands do not suffer from multiple-testing problems and are guaranteed to cover all $ATT(g, t)$ ’s with a probability at least $1 - \alpha$. Finally, we note that our proposed bootstrap procedure can be readily modified to account for clustering, see Remark 10 below.

To proceed, let $\widehat{\Psi}_{t \geq (g-\delta)}^{dr,nev}(W)$ denote the sample-analogue of $\Psi_{t \geq (g-\delta)}^{dr,nev}(W)$, where population expectations are replaced by their empirical analogue, and the true nuisance functions and their derivatives are replaced by their estimators. Let $\{V_i\}_{i=1}^n$ be a sequence of *iid* random variables with zero mean,

unit variance, and finite third moment, independent of the original sample $\{W_i\}_{i=1}^n$. A popular example involves *iid* Bernoulli variates $\{V_i\}$ with $P(V = 1 - \kappa) = \kappa/\sqrt{5}$ and $P(V = \kappa) = 1 - \kappa/\sqrt{5}$, where $\kappa = (\sqrt{5} + 1)/2$, as suggested by [Mammen \(1993\)](#).

We define $\widehat{ATT}_{t \geq (g-\delta)}^{*,dr,nev}$, a bootstrap draw of $\widehat{ATT}_{t \geq (g-\delta)}^{dr,nev}$, via

$$\widehat{ATT}_{t \geq (g-\delta)}^{*,dr,nev} = \widehat{ATT}_{t \geq (g-\delta)}^{dr,nev} + \mathbb{E}_n \left[V \cdot \widehat{\Psi}_{t \geq (g-\delta)}^{dr,nev}(W) \right]. \quad (4.6)$$

The next theorem establishes the asymptotic validity of the multiplier bootstrap procedure proposed above.

Theorem 3. *Under the assumptions of Theorem 2*

$$\sqrt{n} \left(\widehat{ATT}_{t \geq (g-\delta)}^{*,dr,nev} - \widehat{ATT}_{t \geq (g-\delta)}^{dr,nev} \right) \xrightarrow{*} N(0, \Sigma),$$

where Σ is as in Theorem 2, and $\xrightarrow{*}$ denotes weak convergence (convergence in distribution) of the bootstrap law in probability, i.e., conditional on the original sample $\{W_i\}_{i=1}^n$. Additionally, for any continuous functional $\Gamma(\cdot)$,¹⁴

$$\Gamma \left(\sqrt{n} \left(\widehat{ATT}_{t \geq (g-\delta)}^{*,dr,nev} - \widehat{ATT}_{t \geq (g-\delta)}^{dr,nev} \right) \right) \xrightarrow{*} \Gamma(N(0, \Sigma)).$$

We now describe a practical bootstrap algorithm to compute studentized confidence bands that cover $ATT(g, t)$ simultaneously over all $t \geq g - \delta$ with a pre-specified probability $1 - \alpha$ in large samples. This is similar to the bootstrap procedures used in [Kline and Santos \(2012\)](#), [Belloni et al. \(2017\)](#) and [Chernozhukov et al. \(2018\)](#) in different contexts.

Algorithm 1. 1) Draw a realization of $\{V_i\}_{i=1}^n$. 2) Compute $\widehat{ATT}_{t \geq (g-\delta)}^{*,dr,nev}$ as in (4.6), denote its (g, t) -element as $\widehat{ATT}^*(g, t)$, and form a bootstrap draw of its limiting distribution as

$$\hat{R}^*(g, t) = \sqrt{n} \left(\widehat{ATT}^*(g, t) - \widehat{ATT}(g, t) \right).$$

3) Repeat steps 1-2 B times. 4) Compute a bootstrap estimator of the main diagonal of $\Sigma^{1/2}$ such as the bootstrap interquartile range normalized by the interquartile range of the standard normal distribution, $\widehat{\Sigma}^{1/2}(g, t) = (q_{0.75}(g, t) - q_{0.25}(g, t)) / (z_{0.75} - z_{0.25})$, where $q_p(g, t)$ is the p th sample quantile of the $\hat{R}^*(g, t)$ in the B draws, and z_p is the p th quantile of the standard normal distribution. 5) For each bootstrap draw, compute $t\text{-test}_{t \geq (g-\delta)} = \max_{(g, t)} \left| \hat{R}^*(g, t) \right| \widehat{\Sigma}(g, t)^{-1/2}$. 5) Construct $\widehat{c}_{1-\alpha}$ as the empirical $(1 - \alpha)$ -quantile of the B bootstrap draws of $t\text{-test}_{t \geq (g-\delta)}$. 6) Construct the bootstrapped simultaneous confidence band for $ATT(g, t)$, $t \geq (g - \delta)$, as $\widehat{C}(g, t) = [\widehat{ATT}_{dr}^{nev}(g, t; \delta) \pm \widehat{c}_{1-\alpha} \widehat{\Sigma}(g, t)^{-1/2} / \sqrt{n}]$.

The next corollary to Theorem 3 states that the simultaneous confidence band for $ATT(g, t)$ described in Algorithm 1 has correct asymptotic coverage.

Corollary 1. *Under the assumptions of Theorem 2, for any $0 < \alpha < 1$, as $n \rightarrow \infty$,*

$$P \left(ATT(g, t) \in \widehat{C}(g, t) \quad \forall t \in \{2, \dots, \mathcal{T}\}, g \in \mathcal{G}_\delta : t \geq g - \delta \right) \rightarrow 1 - \alpha,$$

¹⁴Since the number of periods $\mathcal{T}$ is fixed, $\Gamma(\cdot)$ should be interpreted as a continuous functional between Euclidean spaces.